

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 2

What Numbers Actually Are

Parallel Reading

Self-Study Curriculum

Parallel Reading: PCM, section I.3 (“Some Fundamental Mathematical Definitions,” subsections on number systems: natural numbers, integers, rationals, reals, complex numbers, pp. 16-30). Read this section carefully: it is dense but short. Then, after the lecture’s section on the sizes of infinity, read PCM section III.11 (“Countable and Uncountable Sets”), two pages.

Also read: Lockhart, *Measurement*, the opening section (“Reality and Imagination”) and the first third of Part One (“On Size and Shape”): roughly the first 70 pages. Lockhart is building intuition about what it means to *measure*, which is secretly a question about what numbers are *for*.